

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Jabaraj T(192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Analytical Problem Solving

- 1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.**

1. Introduction

- Definition of the Water Jug Problem
- Objective of the problem

2. Problem Formulation

- Initial State
- Goal State
- State Space
- Operators (Fill, Empty, Pour)

3. Solution Using State Space Search

- Step-by-step solution
- State transition table
- State-space diagram

4. Search Strategy

- Breadth-First Search (BFS)
- Why BFS is suitable

5. Applications

- Robotics
- Resource allocation
- AI planning

6. Conclusion

- Final state
- Importance of the Water Jug Problem in AI

- 2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.**

- i. What are the percept's for this agent?**
- ii. Characterize the operating environment.**
- iii. What are the actions the agent can take?**

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent? Why?

1. Introduction

- What is a Mars Rover?
- Role of AI in Mars exploration

2. Percepts and Operating Environment

- Percepts
- Environment characteristics (Partially Observable, Dynamic, Sequential, etc.)

3. Actions and Performance Measures

- Actions performed
- Performance evaluation

4. Agent Architecture

- Utility-Based Learning Agent
- Reason for selecting it

5. PEAS Representation

- Performance Measure
- Environment
- Actuators
- Sensors

6. Conclusion

- Importance of AI in autonomous space exploration

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

1. Introduction

- Definition
- Objective of the problem

2. Problem Formulation

- Initial State
- Goal State
- State Space
- Actions
- Path Cost

3. Search Strategy

- Backtracking Search

- Working procedure
- Algorithm
- 4. Solution Representation**
 - State-space tree
 - Example solution
 - Chessboard representation
- 5. Applications**
 - Constraint Satisfaction Problems
 - Scheduling
 - Robotics
 - Optimization
- 6. Conclusion**
 - Importance of Backtracking in AI

- 4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.**

1. Introduction

- OLA as an Intelligent System
- AI in cab booking

2. Goal-Based Agent

- Definition
- Why it is suitable

3. Problem Formulation

- Initial State
- Goal State
- State Space
- Actions

4. Pseudocode and Working

- Pseudocode
- Step-by-step workflow
- Flowchart (optional)

5. Applications and Advantages

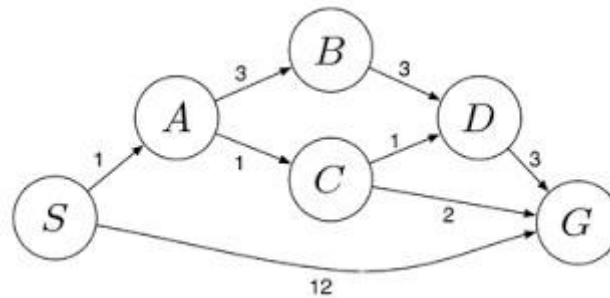
- Ride-sharing
- GPS navigation
- Logistics
- Customer satisfaction

6. Conclusion

- Why Goal-Based Agents are effective for cab booking

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



1. Introduction

- Definition of Uniform Cost Search
- Purpose of UCS

2. Problem Formulation

- Initial State
- Goal State
- State Space
- Path Cost

3. UCS Algorithm and Execution

- Algorithm
- Priority Queue
- Step-by-step node expansion

4. Least-Cost Path

- Search tree
- Cost calculation
- Optimal path

5. Advantages and Applications

- Route optimization
- Logistics
- GPS navigation
- Robotics

6. Conclusion

- Importance of UCS in finding optimal solutions

These **6-subtopic structures** are ideal for **20-mark AI assessment answers** because they are organized, comprehensive, and easy for examiners to evaluate.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
		requirements accurately			